

M.Tech Project / Research Synopsis

Master's Research Project Phase - I
Synopsis Report

Name of College	Ashokrao Mane Group of Institutions
Name of Course	M.Tech. Computer Engineering
Name of Student (PRN No.)	MR. UTKARSH KHANDU KARALE (PRN No.-25PCS1018)
Roll No.	Roll No.-118
Date of Registration	August 2025
Name of Guide	ASST. PROF. VIKAS PATIL
Type of Master's Research Project	Non-Sponsored
Proposed Topic of Master's Research Project	PhishScope: An Explainable Hybrid Intelligence Framework for Static and Dynamic Phishing Website Analysis

1. Introduction / Background

Phishing websites have evolved into sophisticated cyberattacks that employ dynamic web content, JavaScript execution, multi-stage redirections, cloaking techniques, and rapidly changing infrastructure to evade conventional detection mechanisms. While recent machine learning and deep learning approaches have significantly improved detection accuracy, most existing solutions rely primarily on static website features such as URLs, HTML structure, domain information, and SSL certificate metadata. This research introduces PhishScope, an explainable hybrid intelligence framework that combines multi-source static feature analysis with isolated browser-based dynamic execution to generate transparent, evidence-based risk assessments.

2. Relevance / Motivation

- Static-only analysis often fails to capture malicious runtime behaviors revealed during website execution.
- Many current machine learning detection systems operate as black-box models, providing limited interpretability.
- This lack of transparency restricts practical adoption in real-world cybersecurity environments.
- Modern phishing attacks employ dynamic content, JavaScript obfuscation, and multi-stage redirections that static scanners miss entirely.
- Developing an integrated framework that applies Explainable Artificial Intelligence (XAI) to both static and dynamic telemetry will improve detection reliability.
- An explainable phishing detection system will foster user trust and empower cybersecurity analysts with deployable investigation tools.
- A practical deployment via browser extension and desktop application can bridge the gap between academic research and real-world cybersecurity needs.

3. Literature Review

Sr.	Authors	Title	Venue / Year	Major Contribution
1	C. Newen et al.	Uncertainty Awareness and Trust in	2025 IEEE International Conference on	Studies uncertainty explanations and global/local explanations in

		Explainable AI – On Trust Calibration Using Local and Global Explanations	Data Mining (ICDM)	relation to trust calibration; highlights the importance of aligning perceived trust with model reliability.
2	I. Palaiologou, G. Falekas, A. Karlis	Leveraging Expert Knowledge with Explainable Artificial Intelligence for Electric Machine Fault Diagnosis in Industry 5.0	2025 IEEE Energy Conversion Congress and Exposition (ECCE)	Integrates deep learning, uncertainty quantification, XAI and human-in-the- loop analysis for trustworthy decisions under uncertainty.
3	G. Husnain et al.	ViT-Xplain: A Transparent Deepfake Detector for Consumer Electronics Based on Attention and Explainable AI	IEEE Transactions on Consumer Electronics, 2026	Combines model prediction with attention-based explanations and quantitative explanation/trust measures in a domain-specific deepfake detection system.
4	R. Dakhli et al.	Toward a Quantitative Trustworthy Evaluation of Post-Hoc XAI Feature Importance Maps Using Saliency-Based Occlusion	2024 IEEE/ACS 21st International Conference on Computer Systems and Applications (AICCSA), 2024	Evaluates XAI methods using a quantitative faithfulness procedure based on feature importance and occlusion.
5	A. Hämäläinen, A. Karhinen, R.	Building Trustworthy AI	IEEE Access, 2026	Discusses reliability, explainability and trust

	Viitala	Systems for Condition Monitoring		calibration and emphasizes that explanations cannot make an unreliable model trustworthy.
6	B. Kantz et al.	Input Uncertainty Attribution by Uncertainty Propagation	ICASSP 2025	Introduces uncertainty attribution to the input space and connects uncertainty information with explainability and trustworthy ML.
7	L. Tan, C. Huang, X. Yao	A Faithful and Stable Explanation Method for Image Classification	IEEE Transactions on Emerging Topics in Computational Intelligence, 2025	Introduces SCLIME and explicitly targets fidelity, stability and comprehensibility of local explanations.
8	K. Kalasampath et al.	A Literature Review on Applications of Explainable Artificial Intelligence (XAI)	IEEE Access, 2025	Reviews XAI applications and identifies challenges related to interpretability, transparency, trust and evaluation of explanation methods.
9	C. Amadi, A. Ojo	Building Trustworthy AI in Healthcare	IEEE Access, 2025/2026	Reviews trustworthy AI through transparency/explainability, robustness/safety, bias/fairness and related trust and accountability considerations.
10	A. Grobrügge, N. Mishra, J. Jakubik, G. Satzger	Explainability in AI-Based Applications – A Framework for Comparing Different	2024 IEEE 26th International Conference on Business Informatics	Proposes a framework for comparing explanation techniques and assessing agreement between explanations.

		Techniques		
11	M. M. A. Mohua, M. N. Sakib	Explainable and Uncertainty-Aware CO2 Emissions Forecasting Using Quantile Gradient Boosting, SHAP, and Logistic Trend Classification	2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)	Combines uncertainty-aware prediction and SHAP explanations and demonstrates the growing use of joint uncertainty/XAI approaches.

4. Literature Review Summary

The review of current literature highlights a strong dependency on static URL and HTML features for phishing detection. Most contemporary studies focus on maximizing classification accuracy using models like Random Forest, CNNs, or BERT, but heavily neglect dynamic runtime behaviors (such as JavaScript execution or redirect chains). Additionally, while a few recent papers integrate XAI (like SHAP), they typically do so over limited feature sets rather than a comprehensive, multi-source framework.

5. Problem Statement

Phishing websites have evolved into sophisticated cyberattacks that employ dynamic web content, multi-stage redirections, and rapidly changing infrastructure. Most existing solutions rely primarily on static features and operate as black-box models, providing limited interpretability. Therefore, there is a critical need for an explainable hybrid intelligence framework that combines multi-source static analysis with isolated browser-based dynamic execution and Explainable Artificial Intelligence (XAI) to generate transparent, evidence-based risk assessments for practical cybersecurity deployment.

6. Research Gap

- **Limited Integration of Multi-Source Features:** Current models rarely integrate URL, HTML, Domain, and SSL data into a single pipeline.

- **Insufficient Dynamic Behavioral Analysis:** Modern phishing exploits runtime behaviors that static scanners miss entirely.
- **Lack of Explainable Decision-Making:** Prevalent "black-box" systems offer limited transparency, reducing trust.
- **Absence of Integrated Frameworks:** Focus is often placed solely on classification metrics rather than unified risk assessment systems.
- **Limited Practical Deployment:** Validations are frequently offline, lacking user-oriented tools like browser extensions.

7. Aim

To design and develop PhishScope, an explainable hybrid intelligence framework that integrates multi-source static feature analysis with isolated browser-based dynamic behavioral execution and Explainable Artificial Intelligence (XAI) techniques to generate transparent, evidence-based phishing risk assessments for practical cybersecurity deployment.

8. Objectives

1. To extract and integrate multi-source static website features, including URL, HTML, domain, and SSL certificate information.
2. To design and develop a hybrid intelligence framework combining static feature analysis with dynamic browser-based behavioral analysis.
3. To monitor runtime website behaviors (JavaScript, network requests, DOM changes) within an isolated browser environment.
4. To incorporate Explainable Artificial Intelligence (XAI) techniques (SHAP/LIME) for generating transparent predictions.
5. To evaluate the framework's effectiveness, reliability, and practical applicability using standard cybersecurity metrics.

9. Proposed Methodology

- **9.1 Data Collection**

Malicious and benign website instances will be aggregated from repositories like PhishTank, OpenPhish, and curated benign URL lists. The dataset will include diverse phishing attack types to ensure model generalizability.

- **9.2 Multi-Source Static Feature Extraction**

A multi-layer architecture will collect URL-based features (length, special characters, subdomain count), HTML structural features (form actions, iframe usage, script tags), Domain features (WHOIS data, registration age, DNS records), and SSL certificate metadata (issuer, validity, certificate chain).

- **9.3 Dynamic Behavioral Analysis**

An isolated browser sandbox environment will be developed to safely execute websites and monitor runtime behaviors including JavaScript execution patterns, network request sequences, redirect chains, DOM modifications, and browser API interactions.

- **9.4 Feature Preprocessing**

Collected static and dynamic features will be cleaned, normalized, and encoded. Feature scaling will be applied where required. The dataset will be split into 70% Training, 15% Validation, and 15% Test partitions.

- **9.5 Hybrid ML Classification**

Hybrid machine learning models combining Random Forest and XGBoost will be trained on the processed multi-source feature sets. Performance will be evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics.

- **9.6 Explainable AI (XAI) Module**

SHAP (SHapley Additive exPlanations) will be used as the primary feature-attribution method, with LIME (Local Interpretable Model-agnostic Explanations) for comparative analysis. Both global and local explanations will be generated to identify key contributing features for phishing classification decisions.

- **9.7 Risk Scoring and Alerting**

A composite risk scoring mechanism will integrate classification confidence, dynamic behavior indicators, and XAI feature attributions to produce evidence-based risk assessments with categorized threat levels.

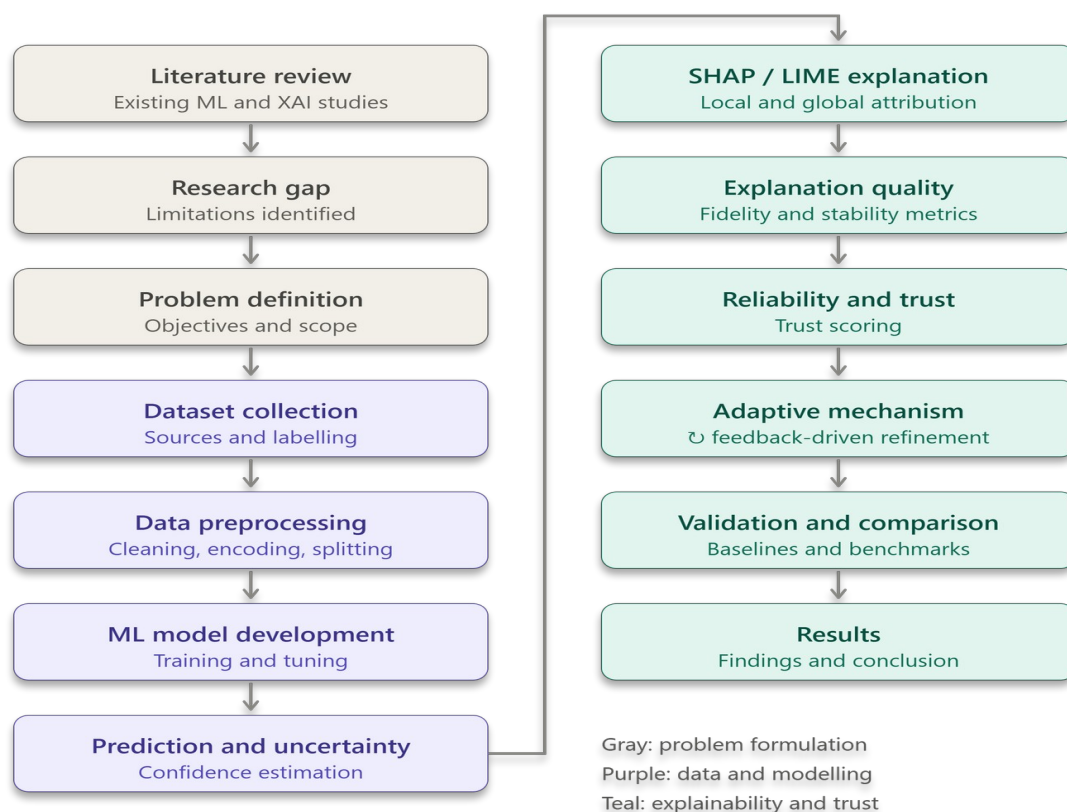
- **9.8 Practical Deployment**

The framework will be deployed through two user-oriented tools: a browser extension for real-time phishing alerts during web browsing, and a desktop investigation application for cybersecurity analysts to perform detailed website analysis.

- **9.9 Validation and Comparison**

The proposed framework will be compared against existing phishing detection methods. Evaluation will include detection performance, explainability quality, runtime efficiency, and practical usability. Cross-dataset experiments will assess generalizability.

10. Methodology Flow



C) Action Plan

Sr. No.	Activity/Month	Aug 26	Oct 26	Dec 26	Mar 27	May 27	Jun 27
		Sept 26	Nov 26	Jan 27	Apr 27		

				Feb 27			
1	Problem Statement Identification						
2	Collecting related data & Information						
3	Design & Analysis						
4	Manufacturing Experimental Setup						
5	Performance Testing, Reading and Calculations						
6	Report Preparation & Patent/Paper Publication						

D) Facilities Available and Requirement

- Computer laboratory / personal computer systems with internet connectivity.
- Python programming environment with Jupyter Notebook, VS Code or equivalent IDE.
- Python libraries: NumPy, Pandas, Scikit-learn, XGBoost, SHAP, LIME, Matplotlib, Selenium, Playwright, and related packages.
- Browser sandbox / isolated execution environment (Docker containers or virtual machines).
- Access to phishing URL databases: PhishTank, OpenPhish, and curated benign URL datasets.
- Cloud/online computing resources for computationally intensive model training and dynamic website execution.

E) Expected Cost of the Dissertation Work

Sr. No.	Particular	Approx. Cost (Rs.)
1	Cloud Instances / Sandbox Infrastructure	10,000
2	Data Acquisition / API Keys	15,000
3	Software Licensing & Security Tools	2,000
4	Publication / Registration Fees	5,000
	Total (Approx.)	32,000

F) Expected Date of Completion

30 June 2027

10. Expected Outcomes

1. A working PhishScope framework capable of providing evidence-based, highly accurate phishing risk assessments.
2. Multi-source feature extraction pipeline integrating URL, HTML, Domain, SSL, and dynamic behavioral indicators.
3. An isolated browser sandbox module for safe dynamic website execution and behavior monitoring.
4. Hybrid ML classification model with demonstrated high performance (Accuracy, Precision, Recall, F1-Score, ROC-AUC).
5. Transparent and interpretable predictions through SHAP and LIME explanations.
6. A browser extension prototype for real-time phishing alerting during web browsing.
7. A desktop investigation application for cybersecurity analysts.
8. A research paper suitable for submission to a reputed conference/journal, subject to the quality and originality of the final results.

11. Research Contribution

- Novel hybrid architecture integrating multi-source static telemetry with isolated browser-based dynamic behavioral analysis for comprehensive phishing detection.
- Integration of Explainable AI (SHAP/LIME) across both static and dynamic feature spaces, providing transparent and interpretable risk assessments.

- Evidence-based risk scoring mechanism combining classification confidence, dynamic behavior indicators, and feature attributions.
- Prediction-level explainability rather than relying only on aggregate model accuracy.
- Practical deployment through a browser extension and desktop investigation application suitable for direct industry integration.

12. Scope of Work

The scope is strictly bounded to the analysis and detection of phishing websites through static indicators and isolated dynamic execution. The study will include multi-source feature extraction (URL, HTML, Domain, SSL), browser-based dynamic behavior monitoring, hybrid ML classification, and XAI-based interpretation.

The initial scope will not include phishing emails, SMS phishing (smishing), voice phishing (vishing), enterprise-scale threat intelligence, or deep malware reverse-engineering. The primary contribution of the present M.Tech work will be the computational framework, experimental evaluation, and prototype deployment tools.

13. Work Plan

Activity	Proposed Duration
Literature Review	September–October 2026
Problem Definition & Methodology	October–November 2026
Design/Simulation/Experiment	November-December 2026
Analysis & Validation	December 2026-February 2027
Thesis & Research Paper	February 2027- June 2027

14. References

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Sharma, B., et al. (2025). Evaluating the Impact of Feature Engineering in Phishing URL Detection. IEEE Access.

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Wilson, F., et al. (2024). Explainable Machine Learning for Phishing Detection. IEEE Access.

Patel, G., et al. (2024). Deep Learning-Based Phishing Website Detection. IEEE Conference.

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Khan, I., et al. (2024). Machine Learning Based Website Phishing Detection. IEEE Conference.

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PG Student

Guide

PG Coordinator

Head of Department

Director